

9. (2.25 points)

Given the following information:

Unlimited Age-to-Age Reported Loss and ALAE
Development Factors

<u>12-24</u>	<u>24-36</u>	<u>36-48</u>	<u>48-60</u>	<u>60-ult</u>
1.463	1.309	1.114	1.043	1.019

Limited Severity Relativities

<u>Limit</u>	<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>	<u>60</u>	<u>Ult</u>
\$100,000	0.95	0.93	0.90	0.88	0.86	0.85
\$250,000	0.99	0.97	0.96	0.95	0.94	0.94

Accident Year 2015 limited reported loss and ALAE at 24 months

<u>Limit</u>	<u>24 months</u>
\$100,000	\$3,450,000
\$250,000	\$3,675,000

Calculate the IBNR for the losses in the \$100,000 to \$250,000 layer for accident year 2015 at 24 months using Siewert's direct development approach.

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QUESTION 9	
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE(S): A4
SAMPLE ANSWERS	
<u>Sample 1</u> $LDF_{24-Ult} = 1.309 * 1.114 * 1.043 * 1.019 = 1.55$ $LDF_{24-Ult}^{0 \text{ to } 250,000} = LDF_{24-Ult} * \frac{R_{Ult}^L}{R_{24}^L}, \text{ where RL is at limit 250 000}$ $= 1.55 * 0.94 / 0.97$ $= 1.502$ $R_{24}^{100,000 \text{ to } 250,000} = \text{Limited severity at 100 000 / Limited severity at 250 000}$ $= 0.93 / 0.97$ $= 0.9588$ $R_{Ult}^{100,000 \text{ to } 250,000} =$ $= 0.85 / 0.94$ $= 0.9043$ $XSLDF_{24-Ult}^{100,000 \text{ to } 250,000} = LDF_{24-Ult}^{0 \text{ to } 250,000} * (1 - R_{Ult}^{100,000 \text{ to } 250,000}) / (1 - R_{24}^{100,000 \text{ to } 250,000})$ $= 1.502 * (1 - 0.9043) / (1 - 0.9588)$ $= 3.488$ Reported losses in layer \$100k to \$250k = 3,675,000 – 3,450,000 = 225,000 By direct development approach, $IBNR = 225,000 * (XSLDF - 1)$ $= 225,000 * 2.488$ $= 559,687.5$ <u>Sample 2</u> $LDF_{24-Ult}^{Unlim} = 1.309 * 1.114 * 1.043 * 1.019 = 1.550$ $LDF_{24-Ult}^{100k-250k} = LDF_{24-Ult}^{Unlim} * (R_{Ult}^{250k} - R_{Ult}^{100k}) / (R_{24}^{250k} - R_{24}^{100k})$ $= 1.550 * \frac{(0.94-0.85)}{(0.97-0.93)} = 3.488$ Rept at 24 100k – 250k = 3675k – 3450k = 225,000 $\text{Unlimited IBNR}_{2015} = 225,000 * (3.488 - 1) = 559,800$ <u>Sample 3</u> Unlimited 24-Ult LDF = 1.309 * 1.114 * 1.043 * 1.019 = 1.55 Limited 100k 24-Ult LDF = LDF * R_{Ult} / R₂₄ = 1.55 * 0.85 / 0.93 = 1.4167 Limited 250k 24-Ult LDF = 1.55 * 0.94 / 0.97 = 1.502	

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$$\text{Limited 250k 24-Ult LDF} = \text{Limited 100k LDF} * R_{24}^{100\text{to}250} + \text{XSLDF} * (1 - R_{24}^{100\text{to}250})$$

$$1.502 = 1.4167 * \left(\frac{0.93}{0.97}\right) + \text{XSLDF} * \left(1 - \frac{0.93}{0.97}\right)$$

XSLDF for Layer between 100k and 250k = 3.485

Reported loss in (100k, 250k) layer = 3,675,000 – 3,450,000 = 250,000

$$\begin{aligned}\text{IBNR} &= 225,000 * (3.485 - 1) \\ &= 559,125\end{aligned}$$

EXAMINER'S REPORT

Candidates were expected to calculate an LDF for losses within a certain layer from unlimited LDF's & severity relativities using Siewert's formula. Then they were to apply this LDF directly to incurred losses to derive IBNR for that layer.

Common errors include:

- Neglecting to calculate the LDF limited to \$250k, before applying Siewert's formula to calculate the layer from \$100k to \$250k
- Not deriving and applying an LDF directly to losses in the layer: they developed the layer limited to \$250k and the layer limited to \$100k and took the difference
- Using severity relativities for entire excess \$100k layer or limited \$250k layer, instead of deriving severity relativities for losses specifically between \$100k and \$250k